

THOMAS FELTON

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Data Science · Quantitative Baseball Research & Analytics

EDUCATION

William & Mary, Williamsburg, VA

Expected Graduation: May 2027

B.S. in Data Science with a concentration in Artificial Intelligence | Minor: Chemistry | GPA: 3.79

Honors: Dean's List, 6 semesters

RELEVANT COURSEWORK

Applied Machine Learning | Statistical Data Analysis | Prob & Stats for Scientists | Databases | Data Visualization |
Generative AI | Supercomputing for Science | Applied Linear Algebra & Calc

RELEVANT EXPERIENCE

Student Manager/Offensive Analytics and Advanced Scouting, William & Mary Baseball

Mar 2024 – Present

- Built a semi-automated scouting pipeline in Python, scraping and aggregating opposing pitcher data from Synergy and Trackman to eliminate manual data entry
- Produced integrated scouting reports — combining Google Sheets analytics and video — for 10–20 opponents per season, delivered to both coaching staff and players ahead of each series

Laboratory Assistant, Meldrum Physical Chemistry Lab, William & Mary

Jan 2024 – Present | 800 hours

- Developed an end-to-end MD simulation pipeline (OpenMM, MDAnalysis) to compute and validate T_2 relaxation times, orchestrating 1,000+ SLURM jobs on W&M's HPC via MPI and Bash
- Engineered a validation suite comparing simulated vs. experimental T_2 values and automated publication-quality diffusion and T_2 figures in Matplotlib and SciPy
- First-author manuscript in production: "Computational Modeling of T_2 Relaxation via Single-Sided NMR"

LEADERSHIP EXPERIENCE

Pitching Coach & President, Club Baseball, William & Mary

Aug 2024 – Present

- Provide individualized mechanical and pitch-development coaching and design structured team practice programs; two-time NCBA All-Region Pitcher, Mid-Atlantic North (1st Team and 3rd Team)

PROJECTS

MLB Pitch Quality (Stuff+) Evaluation System | Python, LightGBM, Optuna, SLURM

- Built an independent, production Stuff+ pitch-quality model — a separate pair of LightGBM models per pitch type (7 TrackMan types): a 3-class whiff/foul/in-play classifier and an xwOBACon contact-quality regressor, composed into a per-pitch run-value grade standardized to a 100 scale each season
- Tuned all 14 sub-models independently via staged Optuna search — each with its own feature set (9 base physical features plus type-specific engineered terms; 15–20, no two alike) and hyperparameters — with a later joint feature sweep scored by 5-fold group-by-pitcher CV against a composite convergent- and predictive-validity objective
- Forecasts a pitcher's next-season ERA, wOBA-against, and xwOBA-against better than each stat predicts its own next-year value — and reproduces FanGraphs' industry-standard Stuff+ at Spearman $\rho \approx 0.76$ on four-seamers (0.658 across 3,068 pitcher-seasons)

Free-Agent Contract Valuation Model — OOTP Simulation League | Python, scikit-learn, Monte Carlo

- Built a front-office-style surplus-value model over 21 seasons of OOTP simulation-league data: park-neutralized component stats (power, contact, baserunning, defense, framing), position-specific delta-method aging curves, and a 40,000-iteration Monte Carlo separating true production from luck
- Priced deals with a ridge-regression market model ($R^2 = 0.568$ on 280 held-out signings) feeding a risk-adjusted surplus optimizer

SMT Data Challenge 2026 — Minor League Ball- & Player-Tracking Analytics

- Participated in the SMT Data Challenge, a student analytics competition using anonymized Minor League Baseball ball- and player-tracking data, building a scoreboard-ready project to drive fan engagement per the prompt; details embargoed until finalists are announced October 2026

SKILLS

Programming: Python, R, SQL, Bash, LaTeX

ML & Data Science: PyTorch, Scikit-learn, LightGBM, CatBoost, Statsmodels, SciPy, Pandas, NumPy, Matplotlib, Seaborn

Baseball Analytics: Statcast, pybaseball, MLB Stats API, TrackMan, Synergy, Baseball Savant, xwOBA & run-value modeling

Unsupervised & Dimensionality Reduction: PCA, t-SNE, UMAP

Simulation & HPC: OpenMM, MDAnalysis, SLURM, MPI, TCSH

Tools & Data Formats: Git, Jupyter, Google Sheets, Excel, Cinema4D; JSON/CSV, .npz, .dcd/.pdb/.mol2